

University Interscholastic League
2022 – 2023 Junior High Number Sense Test B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $353 + 51 =$ _____</p> <p>(2) $17 \times 7 =$ _____</p> <p>(3) $84 - 36 =$ _____</p> <p>(4) $120 \div 8 =$ _____</p> <p>(5) $\frac{7}{16} + \frac{1}{4} =$ _____</p> <p>(6) $32 \div 0.8 =$ _____</p> <p>(7) $0.25 \times 44 =$ _____</p> <p>(8) $21^2 =$ _____</p> <p>(9) $8 \times 6 \times 5 =$ _____</p> <p>*(10) $1981 + 2202 + 18 + 2203 =$ _____</p> <p>(11) $1.5 \times 22 =$ _____</p> <p>(12) The mode of 1, 3, 5, 4, 1 and 2 is _____</p> <p>(13) Which is smaller: $\frac{7}{11}$ or 0.64? _____</p> <p>(14) $83 \times 87 =$ _____</p> <p>(15) $24 \times 12 =$ _____</p> <p>(16) $121 - 29 - 21 =$ _____</p> <p>(17) $72^2 =$ _____</p> <p>(18) $7 + 12 + 17 + 22 + 27 =$ _____</p> <p>(19) MMXXI = _____ (Arabic Numeral)</p> | <p>*(20) $222.222\% \times 269 =$ _____</p> <p>(21) $0.25 \times 12 + 0.5 \times 12 =$ _____</p> <p>(22) $423 \times 111 =$ _____</p> <p>(23) If $f(x) = 15 + 2x^2$, then $f(6) =$ _____</p> <p>(24) $4\frac{4}{5} \times 4\frac{1}{5} =$ _____ (mixed number)</p> <p>(25) $3\frac{5}{6} + 2\frac{2}{3} =$ _____</p> <p>(26) $89^2 + 89 =$ _____</p> <p>(27) The cube root of -216 is _____</p> <p>(28) If n is to 12 as 3 is to 4, then $n =$ _____</p> <p>(29) If $16 + 4x$ is 24 then $x =$ _____</p> <p>*(30) $2022 \times 5 - 2023 =$ _____</p> <p>(31) $105 \times 102 =$ _____</p> <p>(32) The sum of the two largest prime numbers less than 60 is _____</p> <p>(33) If $0.75 + 0.5 = n$, the $n^{-1} =$ _____</p> <p>(34) 24 is divisible by how many whole numbers? _____</p> <p>(35) Three-fourths of a square mile = _____ acres</p> <p>(36) 8 percent = _____ (common fraction)</p> <p>(37) The total cost of item, that costs \$100, with a sales tax of $8\frac{1}{2}\%$ is \$ _____</p> |
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- (38) $101 \times 483 =$ _____
- (39) The area of an equilateral triangle with an altitude of 9-cm is $a\sqrt{3}$ and $a =$ _____ cm^2
- *(40) $250\sqrt{16900} =$ _____
- (41) 46 (base 8) = _____ (base 2)
- (42) $4^3 - 3^2 =$ _____
- (43) The GCF of 8, 24 and 18 is _____
- (44) What is the area of a trapezoid with base 14, 16 and height 15? _____
- (45) $13 \text{ (base 4)} \times 2 \text{ (base 4)} =$ _____ (base 4)
- (46) If $15 \div 3 = \mathbf{n} \div 6$, then $\mathbf{n} =$ _____
- (47) What is the length of an edge of a cube with surface area 150? _____
- (48) $0.25 \text{ m}^2 =$ _____ cm^2
- (49) If $15 - 4x < 43$, then $2x >$ _____
- *(50) $25^3 =$ _____
- (51) In the sequence: 1, 3, $\mathbf{a}$, 7, 9, $\mathbf{b}$, 13, . . .
 $\mathbf{a} + \mathbf{b}^2 =$ _____
- (52) What is the 5th term in the sequence 1, 4, 9, . . .? _____
- (53) What is the sum of the interior angles for a convex polygon with 6 sides? _____ $^\circ$
- (54) What is the product of the LCM and GCD of 16 and 24? _____
- (55) What is the length of the side of a rectangle with perimeter 32 and width 5? _____
- (56) What is the radius of a circle with a circumference of 36π ? _____
- (57) If set $\mathbf{A} = \{1, 2, 3, \dots, 19\}$ and set $\mathbf{B} = \{2, 4, 6, \dots, 20\}$, then the number of elements in $\mathbf{A} \cap \mathbf{B}$ is _____
- (58) $(14^2 + 8 \times 9) \div 6$ has a remainder of _____
- (59) The number of edges for a square pyramid is _____
- *(60) $72^2 \times 101 \div 36 =$ _____
- (61) $(2^{-3} + 2^{-1})^{-1} =$ _____
- (62) 880 yards = _____ mile
- (63) 30 feet/min = _____ yards/sec
- (64) 16% of 16 = _____
- (65) $11^7 \div 7$ has a remainder of _____
- (66) If the volume of a right circular cylinder, with radius 5 and length 10 is $\mathbf{k}\pi$, what is $\mathbf{k}$? _____
- (67) The sum of the positive divisors of 17 is _____
- (68) $4! =$ _____
- (69) $6\frac{3}{4} \div \frac{3}{8} =$ _____
- *(70) $4000\pi^2 =$ _____
- (71) $42^2 + 6^2 =$ _____
- (72) $27\frac{3}{11}\% =$ _____ (common fraction)
- (73) $0.1333\dots =$ _____ (common fraction)
- (74) The volume of a right circular cone with height 3 and radius 6 is $\mathbf{k}\pi$, and $\mathbf{k} =$ _____
- (75) The sum of the 7th and 6th triangular numbers is _____
- (76) $20^2 \times 25 =$ _____
- (77) If the shortest distance between the points (0, 6) and ($\mathbf{x}$, 0) is 10, what is $\mathbf{x}$? _____
- (78) $\frac{4!}{2!} + 12 =$ _____
- (79) $\left(9\frac{1}{2}\right)^2 - \left(5\frac{1}{2}\right)^2 =$ _____
- *(80) 11 miles = _____ yards

2022 – 2023 University Interscholastic League Junior High Number Sense Test B – Key

(1) 404	*(20) 568 – 627	(38) 48783	(59) 8
(2) 119	(21) 9	(39) 27	*(60) 13817 – 15271
(3) 48	(22) 46953	*(40) 30875– 34125	(61) $\frac{8}{5}$; 1.6; $1\frac{3}{5}$
(4) 15	(23) 87	(41) 100110	(62) $\frac{1}{2}$; .5
(5) $\frac{11}{16}$; .6875	(24) $20\frac{4}{25}$	(42) 55	(63) $\frac{1}{6}$
(6) 40	(25) $6\frac{1}{2}$; 6.5; $\frac{13}{2}$	(44) 225	(64) 2.56; $2\frac{14}{25}$; $\frac{64}{25}$
(7) 11	(26) 8010	(45) 32	(65) 4
(8) 441	(27) -6	(46) 30	(66) 250
(9) 240	(28) 9	(47) 5	(67) 18
*(10) 6084 – 6724	(29) 2	(48) 2500	(68) 24
(11) 33	(30) 7683 – 8491	*(50) 14844 – 16406	(69) 18
(12) 1	(31) 10710	(51) 126	*(70) 37505 – 41452
(13) $\frac{7}{11}$	(32) 112	(52) 25	(71) 1800
(14) 7221	(33) $\frac{4}{5}$; .8	(53) 720	(72) $\frac{3}{11}$
(15) 288	(34) 8	(54) 384	(73) $\frac{2}{15}$
(16) 71	(35) 480	(55) 11	(74) 36
(17) 5184	(36) $\frac{2}{25}$	(56) 18	(75) 49
(18) 85	(37) 108.50	(57) 9	(76) 10000
(19) 2021		(58) 4	(77) 8
			(78) 24
			(79) 60
			*(80) 18392 – 20328

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.